

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Derive $(B-a)(B-b) = B(B-a-b) + ab$.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)
- Why does the trick work? Let's see with small numbers!

Show 9×8 using the rule: - 9 is $10-1$ (deficit 1). - 8 is $10-2$ (deficit 2). - Left: $9 - 2 = 7$. Right: $1 \times 2 = 2$. Combined: **72**. - Check: $9 \times 8 = 72$. ✓

“*Same answer as tables! The trick is just another way.*”

Activity (8 min)

A simplified version of *Method Speed Race* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.

- If energy is low, swap the activity for free drawing.